

Identity Elements Abstract Algebra Report

How could identity be “identified”?

In the world, as humans, we have had our identity since we were born and it keeps with us for our entire life. We need it but sometimes we make it as usual and ignore it by ourselves. However, identity is the element that others could recognize us efficiently. Even though identity is vague and various, we can not lose it. The interesting thing is, we also have “identity element” in abstract algebra.

In the world of mathematics, the identity element is a true equality no matter what the values it chose for its variables. When an identity element is applied by a binary operation to another mathematical object, this object leaves unchanged! Are you feeling ridiculous about what I just said? If things are unchanged, why do we need that? No worries, ,actually we can treat the identity element as a non-effect effector. Let’s see and explore together!

Each numeric set can only have one identity element for the binary operation. It is impossible that there exist two or more identity elements in the same binary operation. Let’s see a counter example below,

If for the operation $*$ in X there are two identity elements e and g ,

$$e * g = e \text{ (} g \text{ is the identity element)}$$

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$$e * g = e = g$$

This shows the identity element is unique for the binary operation $*$ in the set X .

In order to own an identity element, the binary operation has to be commutative. There does not exist an identity element in subtraction since subtraction is not commutative.

In a subtraction,

$$x - 0 = x, \text{ however, } 0 - x = -x$$

$$x - 0 \neq 0 - x$$

Let’s see how to identify the identity elements of addition and multiplication below.

Suppose A is a group closed under the binary operation of addition, denoted $+$. e is the additive element of A . The additive identity formula can be written as $(e)+a = a = a+(e)$, $a \in A$. Group A could be real numbers, integers, complex numbers, rational numbers and natural numbers.

Let's try a simple example below!

$$\text{ex. } 6+0 = 6$$

0 is the additive identity element for addition because 0 added to any real number is the original number. The reason why 6 remains unchanged is because there is an additive identity (0) added within this operation.

Suppose B is a group closed under the binary operation of multiplication, denoted $*$. e is the multiplicative element of B . The multiplicative identity formula can be written as $(e)*b = b = b*(e)$, $b \in B$. Group B could be real numbers, integers, complex numbers, rational numbers and natural numbers.

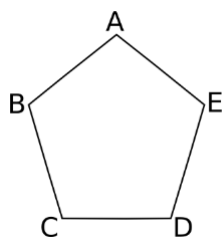
Let's try one more simple example below!

$$\text{ex. } (6+5i)*1 = 6+5i$$

1 is the multiplicative identity element for multiplication because 1 multiplied to any complex number is the original number. The reason why $6+5i$ remains unchanged is because there is a multiplicative identity (1) multiplied within this operation.

Now, you might have a basic idea about identity elements. Let's do a challenge!
Do you think rigid symmetries such as a pentagon has its identity?

$$r^5 = e, j^2 = e$$



If the pentagon is rotated by $2\pi/5$ counterclockwise each time in 5 times, the pentagon stays the same. Same as flipping the pentagon twice. The theory could be applied with other symmetry groups well. Therefore, the identity element of the symmetry group is e .

